NFPA 409-26 AIRCRAFT HANGARS

2016 Edition

Chapter 11 Inspection, Testing, and Maintenance

11.1 Fire Protection Systems.

11.1.1 Inspection, testing, and maintenance of fire protection systems in aircraft hangars shall be performed in accordance with NFPA 11, NFPA 25, *NFPA 70*, *NFPA 72*, or NFPA 80 as applicable and as supplemented by Table 11.1.1.

11.1.2 The 5-year discharge test for piping required in Table 11.1.1 shall not be required if an internal inspection in accordance with NFPA 25 indicates that the pipe is in good condition and free of obstructions, mechanical damage, leakage, and corrosion.

11.1.3 Records of inspections, tests, and test results shall be maintained.

Table 11.1.1 Inspection and Testing of Hangar Fire Protection Systems

System Components	Type and Frequency of Inspections and Tests					
	Weekly	Monthly	Quarterly	Semi-Annually	Annually	Every5 Years
Sprinkler heads	—	—	—	—	V	—
Piping	—	—	—	—	V	D
Pipe hangers	—	—	—	—	V	—
Sprinkler alarm valve	—	V	O1	—	—	—
Deluge valve	—	V	—	—	O	D
Pre-action system	—	V	—	—	D	—
Dry pipe systems	—	V	—	—	D	—
Shutoff valves	—	V	—	—	F	—
Fire pumps	F2	—	—	—	D	—
Water reservoirs	—	V	—	—	—	—
Hose stations	—	V	—	—	—	D
Strainer filter baskets	—	—	—	—	V	—
Foam concentrate	—	—	—	—	F	—
Concentrate storage tanks	—	V	—	—	—	—
Concentrate pumps	F2	—	—	—	O	D
Concentrate control valve (automatic)	—	V	—	—	O	D
Concentrate shutoff valve	—	V	—	—	F	—
Foam proportioning device	—	V	—	—	—	D
Water-powered monitor nozzle	—	V	—	—	D	—
Electric-powered monitor nozzle	—	V	—	—	F	D
Water-powered high-expansion-foam (HEF) Generator	—	V	—	—	D	D
Electric-powered high-expansion-foam (HEF) Generator	—	V	—	—	F	D
Pneumatic detector	—	—	—	F	O3	—
Electric detector	—	—	—	F	O3	—
Optical detector	V	—	—	F	O3	—
Control panels	—	V	—	F	O	—
Alarm transmission (local and remote)	—	F	—	—	—	—
Tamper switch (supervisory switch valve)	—	—	F	—	—	—
Flow indication switch	—	—	—	—	O	—
Low air pressure supervisory switch	—	—	—	F	O	—
Supervisory alarms	—	—	—	F	—	—
Manual actuation stations	—	—	—	F	—	—
Hangar floor drain system and separators	—	V	—	—	—	D
Fire doors	—	V	—	—	F	—
Gas detectors	—	V	—	F	—	—
Ventilation system in pits, tunnels, and ducts	—	—	—	F	—	—
Grounding equipment	—	—	—	—	—	F

V: Visual inspection. D: Operational test with actual discharge. O: Operational test with flow, no discharge. F: Functional test, no flow.
1For the purposes of this test, the inspector's flow valve is acceptable.
2Churn test.
3At this time it is necessary to check that the set points are the same as the original.

Table 11.1.1 Inspection and Testing of Hangar Fire Protection Systems

Type and Frequency of Inspections and Tests

System Components	Every 5 Years
Piping	D: Operational test with actual discharge.
Deluge valve	D: Operational test with actual discharge.
Hose stations	D: Operational test with actual discharge.
Concentrate pumps	D: Operational test with actual discharge.
Concentrate control valve (automatic)	D: Operational test with actual discharge.
Foam proportioning device	D: Operational test with actual discharge.
Electric-powered monitor nozzle	D: Operational test with actual discharge.
Water-powered high-expansion-foam (HEF) Generator	D: Operational test with actual discharge.
Electric-powered high-expansion-foam (HEF) Generator	D: Operational test with actual discharge.
Hangar floor drain system and separators	D: Operational test with actual discharge.
Grounding equipment	F: Functional test, no flow.

V: Visual inspection. D: Operational test with actual discharge. O: Operational test with flow, no discharge. F: Functional test, no flow.
1For the purposes of this test, the inspector's flow valve is acceptable.
2Churn test.
3At this time it is necessary to check that the set points are the same as the original.